Learning in Games

Markov Games, POSGs, Extensive Form

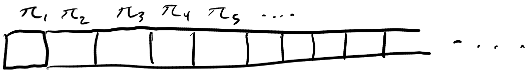
$\underbrace{\hspace{10em}}$

Dynamic Games

MDP as a Normal-Form Game

$\pi_1\ \pi_2\ \pi_3\ \pi_4\ \pi_5\ \dots$

Nature



$|A|^{|S|}$ policies

Dynamic Game in Normal Form

P2

$\pi^2_1, \pi^2_2, \pi^2_3 \dots$

P1

π^1_1

π^1_2

π^1_3

$\vdots$

Classes of games where it's "easy" to find a NE

1 Zero-sum

Optimization problem for finding NE

$$\underset{x,y,u,v}{\text{minimize}} \quad u - \cancel{x^T A y} + v - \cancel{x^T B y}$$

subject to

$$u \geq \hat{x}^T A y \quad \forall \text{ one-hot vectors } \hat{x}$$
$$v \geq x^T B \hat{y} \quad \forall \text{ '' '' '' } \hat{y}$$

Best response constraints

$$\sum_i x_i = 1$$
$$x_i \geq 0$$
$$\sum_i y_i = 1$$
$$y_i \geq 0$$

Zero-sum: $B = -A$

Solving a zero sum game is a LP

2 Potential Games

$\exists\ \psi(a^1, a^2)$ defined so that if a player switches actions, their change in utility is equal to the change in ψ

↑potential function

Examples:

Cooperative
Alligned

2, 2	1, 1
1, 1	3, 3

$A = B$

Prisoner's Dilemma

	coop	defect
coop	-4, -4	-10, -1
defect	-1, -10	-9, -9

$$\psi = \begin{bmatrix} 0 & 3 \\ 3 & 4 \end{bmatrix}$$

Iterative Best Response (IBR) will find a NE in a potential game.

Any game can be decomposed into a potential game + zero-sum game

$$\underset{\text{hard}}{G} = \overset{\leftarrow \text{potential}}{\underset{\substack{\frac{A+B}{2} \\ \text{easy}}}{\bar{G}}} + \underbrace{G - \bar{G}}_{\substack{\text{zero-sum} \\ \text{easy}}}$$